MIL-P-288B
12 May 1982
SUPERSEDING
MIL-P-288A
14 September 1961
MIL-C-2719C
7 February 1962
(See 6.4)

MILITARY SPECIFICATION

PROTRACTORS, ONE OR THREE ARM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers protractors with one or three movable arms used in position, plotting, and hydrographic surveying.

1.2 Classification. protractors shall be of the following types and classes as specified (see 6.2.1):

Type I - Metal
Class 1 - Three arm, left hand, circular
Class 2 - Three arm, right hand, circular
Type 11 - Plastic
Class 1 - Three arm, circular
Class 2 - One arm, semi-circular

20 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3242, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6675

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

L-P-380	- Plastic Molding Material Methacrylate.
L-P-535	- Plastic Sheet (Sheeting): Plastic Strip: Poly (Vinyl Chloride) and Poly (Vinyl Chloride-Vinyl Acetate), Rigid.
GG-M-95	- Magnifier and Reducing Glass.
QQ-B-626	- Brass, Leaded and Nonleaded: Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar and Strip).
QQ-B-750	- Bronze, Phosphor; Bar, Plate, Rod, Sheet, Strip, Flat Wire, and Structural and Special Shaped Sections.
QQ-C-585	- Copper-Nickel-Zinc Alloy Plate, Sheet, Strip, and Bar (Copper Alloy Numbers 735, 745, 752, 762, 766 and 770).
QQ-C-586	- Copper-Nickel-Zinc Alloy; Rod, Shapes and Flat products with Finished Edges (Flat Wire, Strip, and Bar).
QQ-N-290	Nickel Plating (Electrodeposited).
QQ-S-763	- Steel Bars, Wire, Shapes, and Forgings, Corrosion-Resisting.
QQ-S-766	- Steel Plates, Sheets, and Strip-Corrosion Resisting.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-1752	- Cushioning Material, Packaging, Unicellular polyethylene Foam, Flexible.

MILITARY

MIL-P-116	- preservation-packaging, Methods of.
MIL-E-16400	- General Specification for Electronic, Interior Communication, and Navigation Equipment Naval Ship and Shore.

STANDARDS

MILITARY

MIL-STD-105	- Sampling procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.

(Copies of specifications and standards, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Description.

3.1.1 Three-arm protractors. Three-arm protractors shall consist essentially of a circular protractor, graduated in degrees, to which is attached one fixed arm and two movable arms. The movable arms shall be clamped at any angle to the fixed arm, within the limits of the instrument. On type I class 1 instruments, the left hand arm shall be set to small angles down to zero. On type I, class 2 instruments, the right hand arm shall be set to small angles down to zero. On type II class 1 instruments, both movable arms shall be set to small angles down to zero.

3.1.2 One-arm protractor. One-arm protractor shall consist of a semicircular protractor, graduated in degrees, to which is attached one movable arm. The arm shall be clamped at any angle within the limits of the protractor.

3.1.3 Basic specification. In addition to the requirements specified herein, protractors shall be in accordance with the definitions, operating temperature range (see 3.10), the humidity, salt fog, and fungus characteristics of MIL-E-16400. Whenever a requirement of MIL-E-16400 conflicts with a requirement of this specification, the latter shall govern.

3.2 First article. When specified, the contractor shall furnish sample units for first article inspection (see 4.3 and 6.3).

3.3 Material.

3.3.1 Metal parts. Unless otherwise specified, all metal parts shall be of corrosion-resisting materials or other materials treated to resist corrosion. Contact of dissimilar metals, which may cause electrolytic action, shall be avoided.

3.3.2 Plastic parts. As specified herein, all plastic parts shall be made of a colorless transparent material, conforming to L-P-535, composition B, type I, class 2 or L-M-500, type I or II, class 2 and shall have a high polish finish.

3.3.3 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 Accessories. The following accessories shall be furnished with each type I instrument:

(a) Three extension arms
(b) Three thumbscrews for extension arms
(c) One magnifying glass

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

(d) One screwdriver
(e) Four pivot plugs
(f) One stowage and carrying case

3.5 Construction, type I protractors.

3.5.1 Assembly. The assembled protractor shall consist of a circle frame with an inlaid graduated circle and three arms, one of which shall be fixed and attached to the circle frame and two that shall be movable and attached to a pivot center. A tangent screw with vernier scale shall be attached to each of the two movable arms. A reading glass shall be hinged to a rotatable collar, attached to the pivot center. Figure 1 is an example of an acceptable design of the ribs for the circle frame.

3.5.1.1 Accuracy. The graduations on the body of the type I protractors shall be such that no graduation mark will vary from its true position by more than 2 minutes of arc. The movable arms shall be mounted in the assembled instrument so as to rotate on an axis at the center of the graduated circle, with an accuracy such that the maximum error due to eccentricity shall not exceed 3 minutes of arc. The total error of the assembled protractor shall not exceed 5 minutes of arc.

3.5.2 Circle frame. The circle frame shall be brass, conforming to QQ-B-626 alloy 342. The upper surface of the frame shall be machined from the outside diameter to an inside diameter and depth, to match a graduated circle. A graduated circle (see 3.5.2.1) shall be inlaid flush with the top surface of the circle frame. As an option, instead of graduations on the frame, graduations can be combined on a full width graduated circle made of corrosion resistant steel, conforming to QQ-S-766 class 302 (passivated), and silver soldered to the frame. A numbered scale shall be engraved on the top surface of the frame such that, when assembled with the graduated circle, numerals will appear at every 10 degree graduation starting at 360 degrees at the fiducial point and continuing clockwise in decreasing increments of 10 to 180 degrees, then in increasing increments of 10 to 350 degrees. Numerals shall be 0.070 inch high and shall be filled with black enamel. The base of the numerals shall be toward the center of the circle. The frame shall have a circular groove on the undersurface, to receive the guide pins of the clamps. The center of the circle frame shall be a disc not greater than 1-7/16 inches in outside diameter. It shall be connected to the outer circle of the frame by three radial ribs. The center, ribs, and outer circle of the frame shall be fabrocated as one integral piece and shall form a rigid support for the arms. The center of the circle frame shall be fitted for the pivot center.

3.5.2.1 Graduated circle. The graduated circle shall be nickel silver conforming to QQ-C-585, alloy 752, half hard tempered or of corrosion resistant steel conforming to QQ-S-766 class 302 (passivated). The graduated circle shall be not less than 1/2-inch wide, with the outside diameter not less than 6-5/16 inches and the inside diameter not greater than 5-7/32 inches. The entire circle shall be accurately graduated at the outer circumference in degrees and half degrees. The graduated lines shall have a depth of 0.003 inch minimum to 0.005 inch maximum, shall have a width of 0.005 ± 0.001 inch and shall be filled with waterproof ink or enamel. The length of the graduated

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

lines shall be as follows: 10 degree marks 0.19 inch, 5 degree marks 0.15 inch, 1 degree mark 0.11 inch, and 30 minute marks 0.07 inch. Numerals shall be engraved every 10 degrees in both a clockwise and counter-clockwise direction, starting with "O" at the fiducial point and continuing to 180 degrees. The base of the numerals shall be toward the center of the circle.

3.5.3 Pivot center. The pivot center shall be phosphor bronze conforming to QQ-B-750, composition D. It shall be secured to the center of the circle frame by countersunk corrosion-resistant steel or monel screws. The pivot center shall have a projection above the center disc of the frame for retaining the collar of the reading glass, and a projection beneath the center disc for retaining the collars of the two movable arms. The inside and outside surfaces of the pivot center shall be true cylinders, concentric with the circle of the instrument, and shall be ground to an accurate fit for the collars. The pivot shall be free from imperfections and finished smooth.

3.5.4 Pivot plugs. Four pivot plugs shall be furnished for use in marking positions. They shall be made of phosphor bronze conforming to QQ-B-750 composition D, free from imperfections, and finished smooth. The outside diameter of the plugs shall allow for a sliding fit in the pivot center. The four kinds of plugs shall be as follows:

(a) A cylindrical hollow plug, with a glass or clear plastic bottom, having a hole in the center to receive a pencil point and marked with cross lines at right angles on the bottom plastic surface. The glass or plastic bottom shall be set flush in the hollow plug. The hollow plug shall be painted with white enamel on the inside.

(b) A solid cylindrical plug, with a plunger having a point on its lower end, and fitted with a spring to hold the point clear of the bottom when not pressed from above.

(c) A cylindrical plug with a glass or clear plastic bottom having cross lines on the undersurface, at right angles and intersecting accurately at the center on the bottom surface. The glass or plastic bottom shall be set flush in the hollow plug, to avoid parallax. The hollow plug shall be painted with white enamel on the inside.

(d) A solid cylindrical plug having a short needle point in the center of the base.

3.5.5 Arms. The arms shall be of nickel silver conforming to QQ-C-586, alloy 752, half hard tempered or corrosion resistant steel conforming to QQ-S-766 class 302 (passivated). The arms shall be flat bars having a length of 18 inches minimum to 18-3/4 inches maximum from the center point of the instrument. Each arm shall be l/8-inch in thickness with a top-edge bevel angle of 20 to 45 degrees along the radial straightedge and a face width of 1/32 to 1/16-inch. Each arm shall be 0.6 inch in width at its outer extremity and shall be tapered to accomplish the following construction. The fixed arm shall be secured, by means of nickel silver screws and pins, to a rib of the circle frame in such a position that the beveled edge will lie on the radius

Source: https://assist.dla.mil -- Downloaded: 2026l0-16T06:58Z
Check the source to verify that this is the currentversion before use.

of the circle, which passes through the fiducial point of the graduated circle. The two movable arms shall be secured to collars which rotate about the pivot. center. Each movable arm shall be fitted with a vernier, clamp, and tangent screw. The collars shall be reduced in thickness, so that when the arms are secured in place they will come flush with the bottom of the pivot center and lie in the same plane.

3.5.5.1 Straightedges, type I, class 1. Using the center point as datum, the straightedge of the left movable arm for the type 1, class 1 protractor shall lie on a radius of the circle and shall pass through the fiducial point of the graduated circle, when the left arm of the protractor is closed. The straightedge of the right movable arm for the type 1, class 1 protractor shall lie on a radius of the circle and shall be at an angle not greater than 11 degrees, 30 minutes from the fiducial point, when the right arm of the protractor is closed.

3.5.5.2 Straightedges, type I, class 2. Using the center point as datum, the straightedge of the right movable arm for the type I, class 2 protractor shall lie on a radius of the circle and shall pass through the fiducial point of the graduated circle when the right arm of the protractor is closed. The straightedge of the left movable arm for the type I, class 2 protractor shall lie on a radius of the circle and shall be at an angle not greater than 11 degrees, 30 minutes from the fiducial point when the left arm of the protractor is closed.

3.5.5.3 Angle range, 260 degrees. The straightedge of the left movable arm for the type 1, class 1 protractor and the straightedge of the right movable arm for the type I, class 2 protractor shall be capable of laying off angles up to 260 degrees.

3.5.5.4 Angle range, 270 degrees. The straightedge of the right movable arm for the type I, class 1 protractor and the straightedge of the left movable arm for the type I, class 2 protractor shall be capable of laying off angles up to 270 degrees.

3.5.6 Extension arms. Three extension arms, made of the same material specified in 3.5.5, shall be provided. Each extension arm shall be 13-1/2 inches long, tapering in width from 0.6 to 0.4 inch at the outer end, and shall be joined to the permanent arms by scarfing pieces. The scarfing pieces shall be brazed or riveted to the extension arms and secured, when needed, to the permanent arms by means of a threaded thumbscrew and two dowel pins, affixed to the scarfing pieces. Extension arms shall be matched marked to the mating arm.

3.5.7 Vernier scales. The vernier scales shall be of nickel silver conforming to QQ-C-585, alloy 752, half hard tempered or of corrosion resistant steel conforming to QQ-S-766 class 302 (passivated). The upper surface of the vernier scales shall be flush with the surface of the graduated circle. The vernier of each movable arm shall have its zero graduation positioned over the straightedge of the arm. Each vernier shall embrace 14 degrees, 30 minutes of the graduated circle and shall be so graduated as to permit reading of the scale to 1 minute of arc. Each vernier shall have one additional mark on each end of the scale. The graduation requirements shall be the same as those of the graduated circle (see 3.5.2). The engraved numerals shall be numbered 0, 10, 20, and 30.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

3.5.8 Clamps. Each clamp shall be secured to the circle frame by a thumb-screw, and guided in motion by a tongue which slides in the circular groove in the underside of the circle frame.

3.5.9 Tangent screws. The tangent screws shall be made of corrosion resistant steel conforming to QQ-S-763 class 302 (passivated). They shall be of a standard size #4 UNF 48 threads per inch and shall have knurled heads. Each tangent screw shall be attached to its clamp, by a fixed ball working in a socket joint, and to the movable arm, by a ball nut working in a socket joint.

3.5.10 Reading glass. The reading glass shall be a 3.0X single lens magnifier having a focal length of 1-1/2 inches and a clear aperture of 5/8-inch. It shall be hinged to a collar which rotates around the pivot center, so as to have motion parallel to the plane of the instrument. The field of view of the reading glass, when the glass is parallel to the plane of the instrument, shall be centered over the graduated edges of the graduated circle and the vernier scale.

3.5.11 Magnifying glass. A pocket magnifying glass of 3.0X magnifying power, conforming to GG-M-95, type 1, class A, shall be furnished with each type I protractor.

3.5.12 Screwdriver. A screwdriver of good commercial quality of a size fitting the screws on the instrument, shall be furnished with each type I protractor.

3.6 Construction, type II, class 1 protractor.

3.6.1 Assembly. The assembled protractor shall consist of a body having a graduated circle with a fixed integral arm, a pivot center, two movable arms with vernier scales, and thumb nut clamps on each movable arm (see figure 2).

3.6.1.1 Accuracy. The total error of the assembled protractor shall not exceed 6 minutes of arc.

3.6.2 Body. The body shall be made of plastic 0.040 inch (nominal) thick (see 3.3.2). The body shall be a circle 12-3/16 inches in diameter (plus or minus l/16-inch) with a fixed integral arm extending 13-1/2 inches (plus or minus l/16-inch) from the center of the circle. The arm shall taper from a width of 2-1/2 inches measured on the circumference of the circle to a width of approximately 1 inch at the rounded end. A fiducial line shall extend from the pivot center through the center of the fixed arm for the full length of the arm. Two holes, 3/64-inch in diameter, shall be drilled through the fiducial line, one at 7/16-inch from the outer end and the other at 7 inches from the outer end of the arm. The circle shall be accurately graduated on the upper surface, as specified in 3.6.4.

3.6.3 Movable arms. There shall be two movable arms made of plastic 0.030 inch (nominal) thickness (see 3.3.2), which shall pivot about the center of the body. Both movable arms shall be the same length as the fixed arm and shall be rounded on the outer ends. Each movable arm shall have a fiducial line extending radially from the pivot center through the greatest length of the arm. Each arm shall have a smooth straightedge parallel to and 5/16-inch

Source: https://assist.dla.mil -- Downloaded: 2026l0-16T06:58Z
Check the source to verify that this is the currentversion before use.

distance from the fiducial line. The smooth parallel edge of one arm shall be to the left of the fiducial line and on the other arm, to the right of the fiducial line, when the two fiducial lines are in coincidence. The shape of each arm on the opposite side to the smooth parallel edge shall be such that the arm covers approximately 33 degrees of the protractor graduations, when the fiducial lines of the movable arms are in coincidence with the fiducial line of the fixed arm. This side of the arms shall extend past the periphery of the protractor body a distance of 3/8-inch, to accommodate a clamp (see 3.6.6), and shall taper toward the outer end at such an angle that the total width of the arm at the outer end is approximately 1 inch. Two holes, 3/64-inch in diameter, shall be drilled through the fiducial line, one 7/16 inch from the outer end and the other 7 inches from the outer end of each arm. When the fiducial lines of the two movable arms are in coincidence with the fiducial line of the fixed arm, their holes shall coincide with the holes in the fixed arm. The movable arms shall be accurately graduated on the under side to serve as verniers, as specified in 3.6.4.

3.6.4 Graduations. The protractor body shall be accurately graduated and numbered on the upper surface. The graduations shall be either engraved or printed with a tolerance of plus or minus 5 minutes of arc. The outside diameter of the graduations shall be 11 inches (plus or minus 1/16 inch). The fiducial line of the fixed arm shall form the zero line, and the graduations shall be marked in one degree increments and numbered every 10 degrees. Numerals shall have their base toward the center and shall read from 0 to 350 degrees in two rows of numerals, one in a clockwise and one in a counter-clockwise direction. Clockwise numerals shall form the outside row. The movable arms shall be accurately graduated and numbered on the underside, such that the numerals can be read correctly from the top side. The graduations on the movable arms shall serve as verniers, readable to 2 minutes of arc. They shall be numbered every fifth graduation, starting with zero at the fiducial line and continuing from 10 through 60. The base of the vernier numerals shall be towards the center of the body. The vernier graduations shall be concentric with the circle graduations and shall have their base in coincidence with the outside diameter of the graduated circle. All graduation marks and fiducial lines shall be 0.004 to 0.010 inch wide. The length of the graduation marks shall be as follows: 1-degree and 2 minute marks, 5/32 inch; 5-degree and 10 minute marks, 1/4 inch; 10-degree marks, 5/16 inch. Numerals on the body shall be 5/32 inch high and on the movable arms, 1/8 inch high. All graduation marks, fiducial lines, and numerals shall be filled or printed with permanent waterproof black ink or black enamel. They shall be protected with a transparent plastic cover sheet from 0,003 to 0.005 inch thick of a material compatible with the base material. When tested as specified in 4.4.4, there shall be no evidence of the cover sheet separating.

3.6.5 Pivot center. The pivot center shall consist of a brass center post, knurled ring nut, and plastic washer arranged to allow for free axial movement of the movable arms. The center post and knurled nut shall be made of brass and shall have an overall height not to exceed l/4-inch, when assembled. The bottom of the post shall have a flange 3/8-inch in diameter which shall project 1/32-inch below the underside of the protractor body. The edges of the flange shall be slightly rounded, in order to facilitate sliding of the protractor over the chart surface. The stem of the post shall be 1/4-inch in diameter and shall be threaded to receive the knurled nut. The top of the post shall have an opening approximately 3/16-inch in diameter and counterbored

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

to within l/32-inch of the bottom. A 3/64-inch diameter hole shall be drilled through the bottom center of the post. A plastic washer (see 3.3.2) of 1-1/2-inch diameter shall be provided over the two movable arms. A knurled ring nut shall be provided to secure the washer and movable arms, yet allow for free axial movement of the arms.

3.6.6 Clamps. Each movable arm shall be provided with a "U"-shaped brass clamp. The bottom of each clamp shall have a threaded post secured so that it will not turn. The top of the clamps shall be shaped at the inner edge, so that they will not obstruct the view of any of the numerals or graduations on the vernier. A black molded plastic knob, with a brass thread insert, shall be used for tightening the clamp. A brass washer, 1/32-inch thick, shall be provided between the knob and the clamp.

3.7 Construction, type II, class 2 protractor.

3.7.1 Assembly. The assembled protractor shall consist of a body having a graduated semi-circle with a movable arm containing a vernier scale and thumb nut clamp (see figure 3).

3.7.1.1 Accuracy. The total error of the assembled protractor shall not exceed 6 minutes of arc. The body, arm, and center post shall be assembled so that the beveled edge of the arm coincides radially at all angles with the graduations on the edge of the body. The graduated edge of the arm shall bisect the grid squares diagonally when alined with the 45- and 315-degree graduations, and shall coincide vertically with the line from the radial center to the zero graduation and horizontally with the line through the 90- and 270-degree graduations. The clamping arrangement shall allow the arm to be rotated without friction on the top surface of the body, yet allow for positive locking of the arm at any desired position.

3.7.2 Body. The body shall be plastic (see 3.3.2). The body shall be a semicircular disk, 4.875 ± 0.010 inches high, 4.000 ± 0.005 inches in radius, and 0.080 ± 0.010 inch thick. The center of radius shall be located 7/8-inch above the lower edge of the body on the vertical centerline. A clearance hole and keyway for the center post and a counterbored hole on the underside of the body shall be provided for the center post flange, at the center of radius of the body.

3.7.2.1 Markings. All markings shall be on the underside and protected with a transparent plastic cover sheet from 0.003- to 0.005-inch thick of a material compatible with the body.

3.7.2.2 Graduations. The graduations shall be either stamped or engraved with a tolerance of plus or minus 5 minutes of arc. The circular edge shall be graduated in half degrees with lines 0.004- to 0.010-inch wide. The length of the graduation marks shall be 1/2 degree marks, 1/8-inch; 1 degree marks, 3/16-inch; 5 and 10 degree marks, 5/16-inch. Each 10-degree line shall be designated with numerals 5/64-inch high. The outer row shall be numbered to read from left to right, from 260 to 350 degrees with zero at the top, and continuing up to 100 degrees, The second or inner row shall read from left to right beginning with 80 and continuing to 280 degrees. All markings shall be filled or printed with permanent waterproof ink or enamel.

Source: https://assist.dla.mil -- Downloaded: 2026l0-16T06:58Z
Check the source to verify that this is the currentversion before use.

3.7.2.3 Grid scale. The grid scale shall consist of 1/2-inch squares covering the free area of the body as shown on figure 3. The center vertical line of the grid shall pass through the radial center in line with the 0- and 180-degree markings from the bottom edge of the body to a height of 3-inches above the radial center. All other vertical grid lines shall be parallel to the center vertical grid lines. The center horizontal grid line shall pass through the radial center, in line with the 90 and 270 degree markings for a length of 3-inches on each side of the radial center. All other horizontal grid lines shall be parallel to the center horizontal grid line. Vertical and horizontal grid lines passing through the radial center shall be in alinement with the graduations as indicated, within a tolerance of plus or minus 15 minutes of arc. All other grid lines shall be parallel to the center lines within the same tolerance. The grid lines shall be filled with magenta ink no. X70248, manufactured by Hill-Hentshel Ink Company, St. Louis, MO, or magenta ink no. M21135 manufactured by Sinclair and Valentine Company, Baltimore, MD, or equivalent.

3.7.3 Arm. The arm shall be made of molded plastic (see 3.3.2) in accordance with figure 4. The arm shall contain a beveled edge having a very light matte surface suitable for pencil marking. The arm shall be graduated on the underside of the beveled edge in 1/10-inch increments, with three lengths of lines designating 1/10-, 1/2-, and 1-inch, beginning at a point 4.400 inches from the center post hole and continuing to the end of the arm. The l-inch lines shall be 1/8-inch long, the 1/2-inch line shall be 3/32-inch long, and the 1/10-inch lines shall be 1/16-inch long. Each inch mark shall be numbered beginning at 5 and ending at 17. Numerals shall be 3/32-inch high. The numerals and graduations shall be either engraved or molded in the underside such that they can be read correctly from the top side of the arm. Markings shall be 0.015-inch deep by 0.010-inch wide and filled with permanent waterproof ink or ename1, which shall bond or fuse to the plastic. The arm shall be annealed after molding, to relieve stresses.

3.7.4 Center post assembly. The center post assembly shall be made of brass conforming to QQ-B-626, alloy 342. All parts shall be nickel-plated conforming to QQ-N-290, class 1, grade E. The center post assembly shall consist of a flanged center post with key, threaded retainer rig, keyed washer, and knurled thumb nut. The center post shall be conically drilled so that a sharpened lead pencil may be inserted through the post to the paper under the protractor. The center post stem shall be threaded, approximately 7/16-inch in diameter, with the conical hole 9/32-inch in diameter at the top converging to approximately 3/32-inch in diameter at the bottom. A key shall be firmly fastened to the center post flanged edge by pinning, brazing, or silver soldering to prevent turning of the center post in the body. The center post shall be shouldered and threaded to receive a retainer ring with a threaded hole approximately 19/32-inch in diameter. The retainer ring shall have two holes for fastening with a spanner wrench. The outside diameter of the center post stem shall be threaded to receive the knurled thumb nut. A keyway shall be provided on the stem to receive a keyed washer, which shall prevent the arm from turning when the thumb nut is tightened. The thumb nut shall be approximately 3/4-inch in diameter by 3/8-inch high, with a threaded hole to fit the center post stem. The hole in the top of the thumb nut shall be countersunk approximately 3/32-inch deep. The general configuration of the center post assembly is shown in section “A-A” of figure 3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

3.8 Identification. Each protractor shall be identified in accordance with MIL-STD-130, using permanent waterproof black ink or enamel. Location of identification shall be: type I on the top of a rib of the circle frame; type II on the underside of the body such that it can be correctly read from top; type II class 1 below the pivot point; type II class 2 between the scale numeral and the top of the grid lines.

3.9 Stowage and carrying case.

3.9.1 Type I protractor. The storage and carrying case for type I protractors shall be of hardwood, with locked corners, bottom secured by countersunk brass screws, and a lid secured by inside hinges and closed by hooks and screw-eyes. The case shall be finished smooth and varnished inside and outside. The interior bottom and all bearing surfaces shall be covered with a dark color velvet or other suitable lining material. The covering shall be of fungus-resistant material or shall be treated to resist the growth of fungi. Fittings for securing the instrument and accessories shall be fastened by countersunk brass screws or waterproof glue. The case shall be only large enough to contain the closed instrument and accessories (see 3.4). The accessories shall be housed within the case, to prevent them from becoming loose during transit. All metal hardware shall be of good commercial quality, treated or finished to be corrosion resistant. A metal or plastic case equal to the strength and rigidity of the wood case may be offered for approval by the contracting activity. Each case shall be identified with a corrosion resistant information plate attached to the lower front face of the case and containing the information specified in 3.8.

3.9.2 Type II protractor. Each type 11 protractor shall be provided with a folding type paperboard envelope or folder. The envelope or folder shall be marked with waterproof ink, with the information specified in 3.8.

3.10 Storage temperatures. Type I protractors and cases shall not be damaged by storage temperatures greater than 160°F and less than 80°F. Type 11 protractors, less cases, shall not be damaged by storage temperatures greater than 140°F and less than 80°F.

3.11 Workmanship. Workmanship shall be in accordance with good commercial practice for this type of equipment and be free of warpage, cracks, splits, sharp edges, or burrs.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspections requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of one type I protractor and three type II protractors and shall be subjected to the examination and tests specified in table I for group A, B, and C inspection.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of group A, B, and C tests of 4.4.2, 4.4.3, and 4.4.4.

4.4.1 Lot. For purposes of inspection, all protractors of the same type and class offered for delivery at one time shall be considered a lot.

4.4.1.1 Sampling for group A examination and test. A random sample of protractors shall be selected from each lot in accordance with MIL-STD-105.

4.4.1.2 Sampling for group B examination and test. A random sample of protractors shall be selected from each lot in accordance with MIL-STD-105, inspection level S-1, and subject to the group B tests of 4.4.3.

4.4.1.3 Sampling for group C examination and test. one of type I or three of type II protractors shall be selected at random from any lot and shall be subjected to the group C tests of 4.4.4.

4.4.2 Group A examination and test. Each protractor selected in accordance with 4.4.1.1 shall be subjected to group A tests of table 1. Any protractor failing any group A test shall be rejected. The contractor shall correct the cause of failure on the rejected protractor and all other production units so affected (see 4.4.5).

4.4.3 Group B examination and test. Each protractor selected in accordance with 4.4.1.2 shall be subjected to group B tests of table I. Any protractor failing any group B test shall be rejected. The contractor shall correct the cause of failure on the rejected protractor and all other production units as affected (see 4.4.5).

4.4.4 Group C examination and test. Each protractor selected in accordance with 4.4.1.3 shall be subjected to group C tests of table I. Any protractor failing any group C test shall be rejected. Failure of sample protractor(s) shall be cause for rejection of the lot (see 4.4.5).

4.4.5 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot, rework it to correct the defects, or screen out the defective units, as applicable, and reinspect. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

TABLE I. Quality conformance inspection.

Examination and test	Requirement	Test method	AQL Percent defective MIL-STD-105 major	minor
Group A				
Surface examination	3.3 3.4 3.5	4.5.1	1.5	4.0
	3.6	4.5.1	1.5	4.0
	3.7	4.5.1	1.5	4.0
Group B				
Accuracy	3.5.1.1	4.5.2, 4.5.2.1, 4.5.2.4	.25	1.0
	3.6.1.1	4.5.2, 4.5.2.2, 4.5.2.4	.25	1.0
	3.7.1.1	4.5.2, 4.5.2.3, 4.5.2.4	.25	1.0
Group C				
Storage temperature	3.1.3 and 3.10	MIL-E-16400	see 4.4.1.3	
Humidity	3.1.3	MIL-E-16400	see 4.4.1.3	
Salt fog	3.1.3	MIL-E-16400	see 4.4.1.3	
Fungus	3.1.3	MIL-E-16400	see 4.4.1.3	
Vibration	3.1.3	MIL-E-16400	see 4.4.1.3	

4.5 Examination and test method.

4.5.1 Surface examination. Each protractor selected for test under 4.4.2 shall be visually and dimensionally examined to verify conformance with requirements of this specification. Defects for basis of acceptable quality level are classified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

TABLE 11. Classification of defects.

Category	Defects
Critical:	
1	None defined in this table.
Major:	
101	Incomplete; component parts missing (see 3.4, 3.5.1, 3.6.1, 3.7.1).
102	Material not new, nor as specified (see 3.3).
103	Not free of cracks, splits, sharp edges, or burrs (see 3.11). Circle frame non-conforming (see 3.5.2).
104	Evidence of warpage (see 3.11).
105	Graduated circle width, outside and inside diameter (see 3.5.2.1).
106	Ribs fail to provide support required.
107	Circular groove prevents smooth movement of movable arms; not proper fit for clamp.
108	Center disc diameter (for pivot center).
109	Center, ribs, and outer circle not one integral piece.
110	Graduated circle inlay not secured.
111	Inlay not flush with top surface.
112	Numeral scale not engraved as specified.
113	Graduated circle not engraved as specified.
114	pivot center not securely attached (see 3.5.3).
115	pivot center length, outside and inside diameter non-conforming; not concentric (see 3.5.3); surface finish not as specified (see 3.5, 3.6, and 3.7).
116	pivot plugs fail to fit pivot center as specified; plug glass or point non-conforming (see 3.5.4).
117	Arms, length, width, and thickness not as specified; not flat; not tapered (see 3.5.5).
118	Top edge bevel less than specified; does not provide straight edge (see 3.5.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

TABLE II. Classification of defects. - Continued

Category	Defects
119	Fixed or movable arms not secured as specified (see 3.5.5).
120	Fixed arm not positioned as specified (see 3.5.5.1, 3.5.5.2).
121	Movable arms not free or bound on pivot center (see 3.5.5).
122	Clamp and tangent screw inoperable or excessive looseness; fails to secure in position (see 3.5.8, 3.5.9).
123	Vernier not flush with graduated circle (see 3.5.7).
124	Vernier graduations not as specified; engraving not legible; depth and width of lines non-conforming (see 3.5.7, 3.5.2).
125	Reading glass not magnifying and focal length non-conforming (see 3.5.10).
126	Reading glass not operable; evidence of binding or excessive looseness (see 3.5.10).
127	Accessories incomplete, missing, or not as specified (see 3.4, 3.5.11, 3.5.12).
128	Plastic material thickness less than specified (see 3.6.2, 3.6.3, 3.7.2).
129	Finish non-conforming (see 3.3.2).
130	Body not graduated as specified (see 3.6.4, 3.7.2.2).
131	Fiducial line(s) non-conforming, not straight, not legible (see 3.6.2, 3.6.3).
132	Movable arm(s) straight edge not as required (see 3.6.3, 3.7.3).
133	Movable arm(s) graduations (vernier) not as specified (see 3.6.3).
134	Graduation marks, fiducial lines and numerals not filled with specified material (see 3.6.4, 3.7.2.2, 3.7.2.3, 3.7.3).
135	Pivot center and clamps non-conforming; evidence of binding or excessive looseness (see 3.6.5, 3.6.6, 3.7.4).
136	Hole not counterbored and clearance hole not as specified (see 3.7.4).
Minor	
201	Marking missing, incorrect, or not legible (see 3.8).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

TABLE II. Classification of defects. - Continued

Category	Defects
202	Carrying case, envelope or folder non-conforming (see 3.9). Type II protractor non-conforming.
203	Body radius (see 3.6.2, 3.7.2).
204	Integral arm from center of circle (see 3.6.3, 3.7.3).
205	Arm taper width (see 3.6.3).
206	Fiducial line length (see 3.6.2).
207	Diameter and location of holes (see 3.6.3).
208	Movable arm(s) length (see 3.6.3, 3.7.3).
209	Graduation marks length (see 3.6.4, 3.7.2.2, 3.7.3).
210	Height of numerals on body and movable arms (see 3.6.4, 3.7.2.2).

4.5.2 Accuracy test. For testing the accuracy of the protractors, a metal check plate, approximately 48-inches square in size, shall be used. The metal plate shall be scribed with an accurate system of radial lines extending from the center of the plate to the periphery at 15 degree intervals from 0 through 360 degrees. The radial lines shall be approximately 0.005-inch in width, and shall be accurately scribed so that the angle between the fiducial line or 0-degree line and each radial line shall be accurate to within 1 minute of arc.

4.5.2.1 Accuracy test for type I protractors. The accuracy of the type I protractor with arms fully extended shall be determined as follows:

(a) The left arm of the class 1 protractor or the right arm of the class 2 protractor shall be held against the fixed arm and the graduated circle and vernier shall be read to check the accuracy of the zero setting.

(b) The center of either the class 1 or 2 instruments shall be accurately located at the center of the system of radial lines on the master check plate and the fixed arm shall be located accurately along the 0-degree fiducial line. The left and right arms in turn shall be set at 180 degrees on the vernier and circle, and checked for coincidence with the 180 degree mark on the master check plate.

(c) The fixed arm of the protractor shall then be rotated to a random position on the master check plate, making certain that the protractor is still accurately centered on this plate. The protractor shall be anchored in this positicn by weights

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

or other suitable means. With the protractor oriented as described above, each movable arm shall be rotated to each of the 15-degree radial lines on the check plate falling within the working limits of each arm, and readings on the circle and vernier shall be taken at each of these positions. The angle between any one of these positions of the movable arms and all the other positions as specified above, based on the circle and vernier readings of the protractor, shall be compared with the corresponding angles as indicated on the master check plate in order to determine the accuracy of the instrument.

(d) The protractor shall be reoriented about the center of the master check plate, to bring the fixed arm to a random position approximately 180 degrees to the first position used; then the above tests shall be repeated.

4.5.2.2 Accuracy test for type II class 1 protractor. The accuracy of the type 11 class 1 protractor shall be determined as follows:

(a) The fiducial line of each movable arm in turn shall be placed over the fiducial line of the fixed arm and the graduated circle, and the vernier shall be read to check the accuracy of the zero setting.

(b) The center of the instrument shall be accurately located at the center of the system of radial lines on the master check plate and the fiducial line on the fixed arm shall be located over the fiducial line on the check plate. The right and left arms in turn shall be set at 180 degrees on the vernier and circle, and checked for coincidence with the 180 degree mark on the master check plate.

(c) The fixed arm of the protractor shall then be rotated to a random position on the master check plate, making certain that the protractor is still accurately centered on this plate. The protractor shall be anchored in this position by weights or other suitable means. With the protractor oriented as described above, each movable arm shall be rotated to each of the 15-degree radial lines on the check plate falling within the working limits of each arm, and readings on the circle and vernier shall be taken at each of these positions. The angle between any one of these positions of the movable arms and all the other positions as specified above, based on the circle and vernier readings of the protractor, shall be compared with the corresponding angles as indicated on the master check plate in order to determine the accuracy of the instrument.

(d) The protractor shall be reoriented about the center of the master check plate to bring the fixed arm to a random position, approximately 180 degrees to the first position used; then the above tests shall be repeated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

4.5.2.3 Accuracy test for type II class 2 protractor. The assembled protractor shall be tested for conformance to 3.7, by rotating the arm to any desired setting and tightening the knurled thumb nut. The arm shall then be examined for evidence of slippage, when a line is drawn along the length Of the edge of the arm. The test specified herein shall be repeated in at least two additional settings. Reset to previous settings and check for accuracy of previously drawn line.

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions).

5.1 packaging. packaging shall be level A or C as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Type I. Each carrying case (see 3.9.1) complete with protractor and accessories cushioned and secured in locations provided, shall be unit packaged in accordance with method III of MIL-P-116 as follows:

(a) The carrying case shall be overwrapped with neutral grease-proof, waterproof paper and shall be secured with pressure sensitive tape.

(b) The wrapped carrying case shall be cushioned by placing a l-inch minimum thick sheet of unicellular polyethylene cushioning conforming to PPP-C-1752, type 11, class 1 on all surfaces and individually packaged within a fiberboard box conforming to PPP-B-636; type CF; class, weather-resistant; variety, SW; grade W5C or W6C. Box closure shall be method IV as specified in PPP-B-636.

5.1.1.2 Type II. Each protractor shall be wrapped in either tissue not less than 6 layers in thickness, cellulose crepe wadding, or other non-abrasive material. Each wrapped protractor shall be inserted in a carrying envelope (see 3.9.2) and packaged within a snug-fitted folder conforming to PPP-B-636 type SF; class, domestic; grade 125; style OPF,

5.1.2 Level C. Each carrying case or envelope (see 3.9.1 or 3.9.2) complete with protractor and accessories, where applicable, shall be packaged to assure protection against physical damage during shipment, from the supply source to the first receiving activity.

5.2 Packing. packing shall be level A, B, or C as specified (see 6.2.1).

5.2.1 Level A. Protractor packaged as specified in 5.1 shall be packed in fiberboard boxes conforming to PPP-B-636; class weather-resistant; variety SW; grade W5C or W6C. Box closure shall conform to method V as specified in the appendix to PPP-B-636.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

5.2.2 Level B. Protractors packaged as specified in 5.1, shall be packed in fiberboard boxes conforming to PPP-B-636 class domestic, variety SW minimum, grade 200. Box closure shall conform to method I as specified in the appendix to PPP-B-636.

5.2.3 Level C. The packaged protractor shall be packed to assure acceptance by common carrier and afford protection against physical or mechanical damage during direct shipment, from the supply source to the first receiving activity.

5.3 Marking. In addition to any special markings specified elsewhere in the contract, interior (unit) packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Type I protractors are intended primarily for precision plotting and fixes on distant objects. Type II class 1 protractors are intended primarily for rapid position plotting on comparatively close objects. Type II class 2 protractors are intended primarily for measuring distance and laying out angles in position plotting and general drafting room practice.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of protractor required (see 1.2).
(c) Level of packaging and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one type I protractor and three type II protractors. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents' first article.

6.3.1 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - SH
Air Force - 99

Preparing activity:
Navy - SH
(Project 6675-0719)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

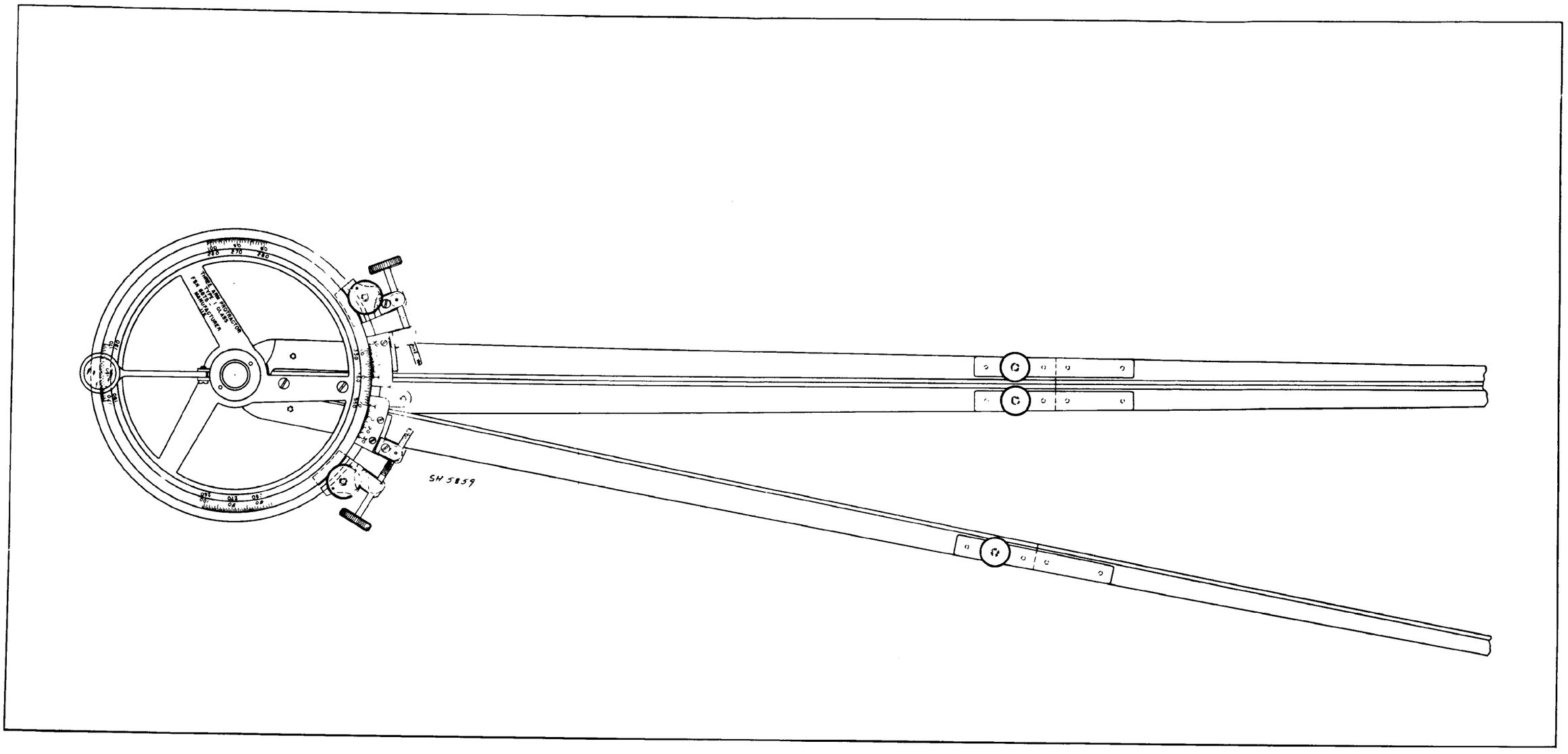


FIGURE 1. Type I protractor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

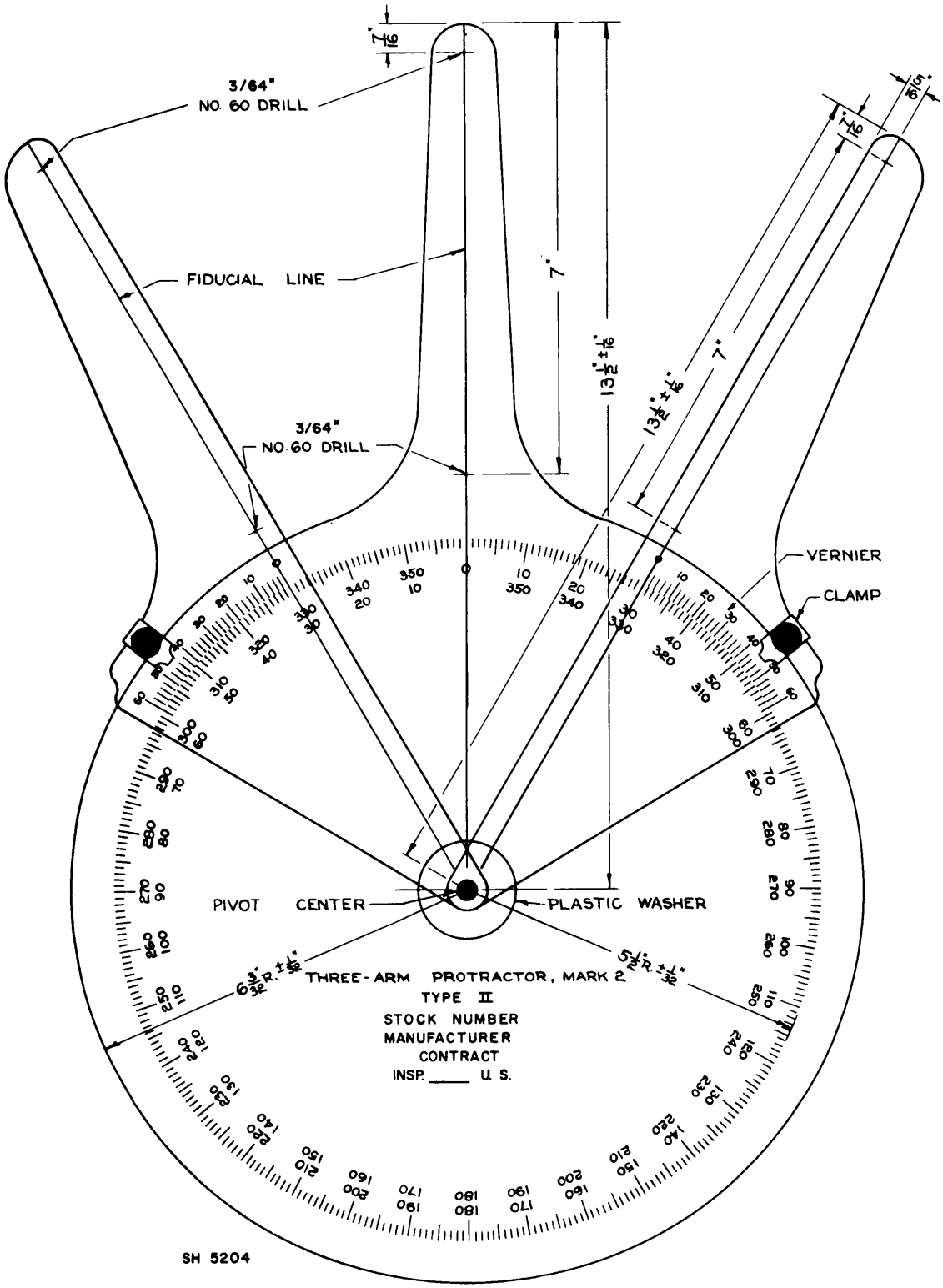


FIGURE 2. Type II class 1 protractor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

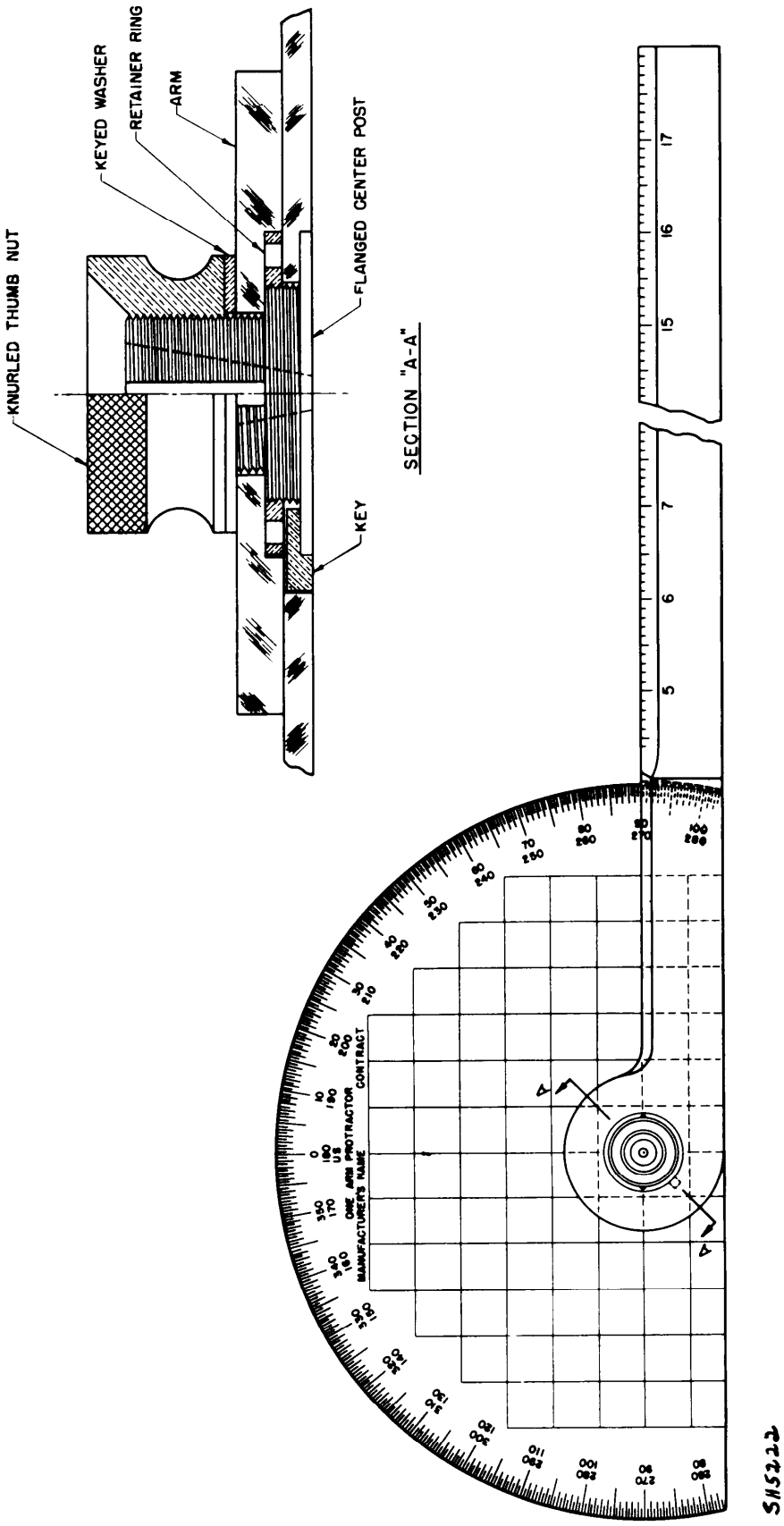


FIGURE 3. Assembly - type II, class 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

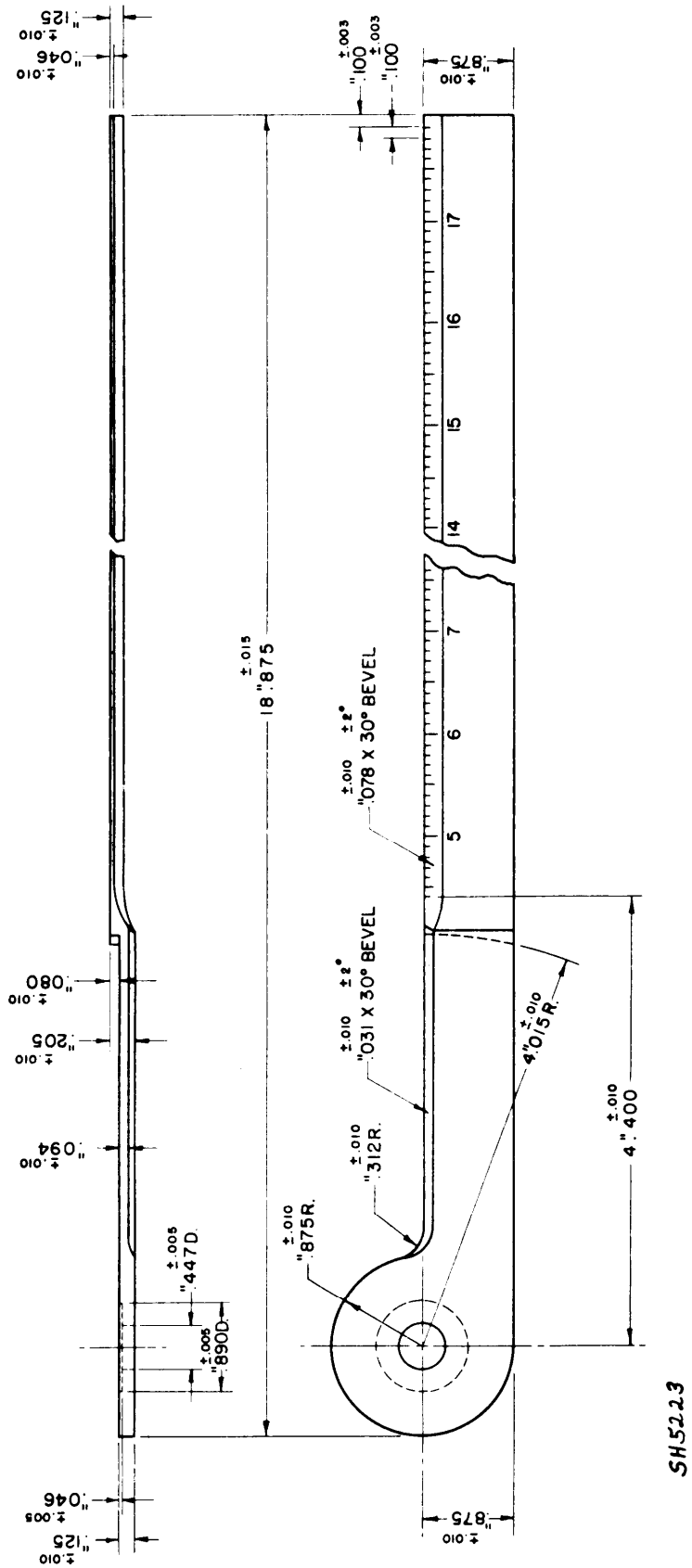


FIGURE 4. Arm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:58Z
Check the source to verify that this is the current version before use.